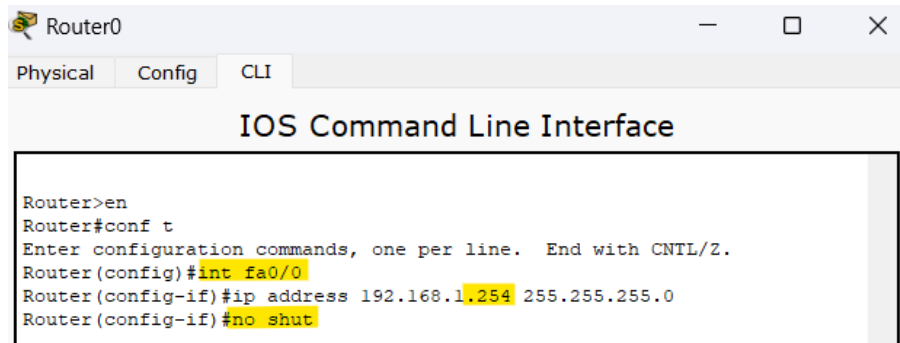


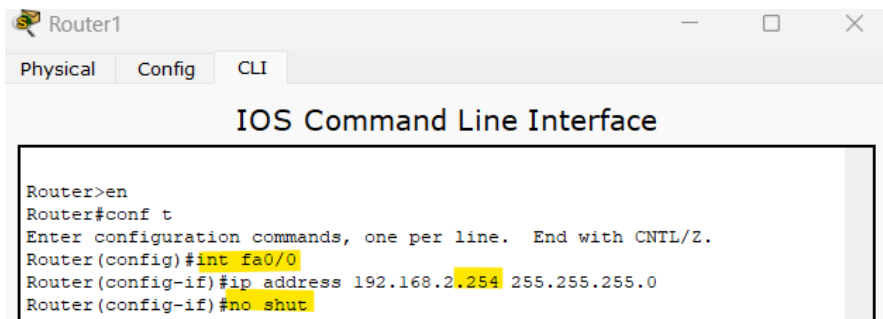
Part 1

(a) Running Configurations of Router 0 and Router 1

A screenshot of the Router0 CLI window. The window has tabs for 'Physical', 'Config', and 'CLI'. The 'CLI' tab is active, showing the 'IOS Command Line Interface'. The command history shows: Router>en, Router#conf t, Router(config)#int fa0/0, Router(config-if)#ip address 192.168.1.254 255.255.255.0, and Router(config-if)#no shut.

```
Router0
Physical Config CLI
IOS Command Line Interface
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip address 192.168.1.254 255.255.255.0
Router(config-if)#no shut
```

Figure 1.1.1 Router0 CLI Showing fa0/0 IP Address Configuration

A screenshot of the Router1 CLI window. The window has tabs for 'Physical', 'Config', and 'CLI'. The 'CLI' tab is active, showing the 'IOS Command Line Interface'. The command history shows: Router>en, Router#conf t, Router(config)#int fa0/0, Router(config-if)#ip address 192.168.2.254 255.255.255.0, and Router(config-if)#no shut.

```
Router1
Physical Config CLI
IOS Command Line Interface
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip address 192.168.2.254 255.255.255.0
Router(config-if)#no shut
```

Figure 1.1.2 Router1 CLI Showing fa0/0 IP Address Configuration

int fa0/0

The fa0/0 interfaces on both routers are configured as the default gateways for their respective internal networks.

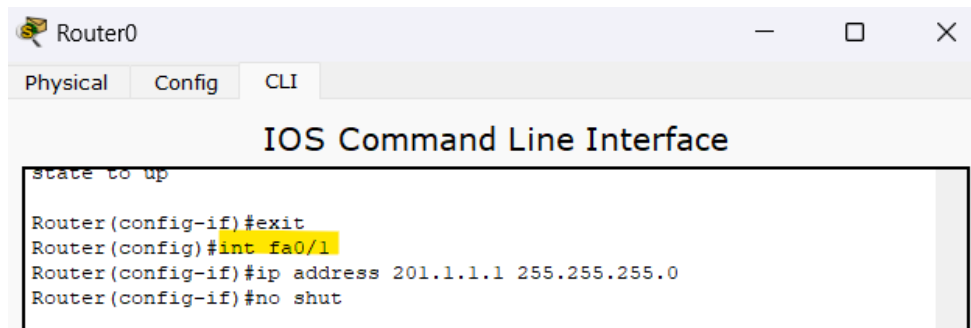
On Router0 (*Figure1.1.1*), the fa0/0 interface is assigned the IP address (192.168.1.254/24) which serves as the default gateway for devices in the internal network (PC0, PC1 & PC2).

On Router1(*Figure1.1.2*), the fa0/0 interface is configured with the IP address (192.168.2.254/24) which acts as the default gateway for devices in the offsite network where the DHCP server resides.

These configurations allow hosts within each subnet to send traffic outside their local network through their respective routers.

.254 – Represents the last usable IP address in each /24 subnet (192.168.1.0 & 192.168.2.0). It is commonly assigned to routers as the default gateway.

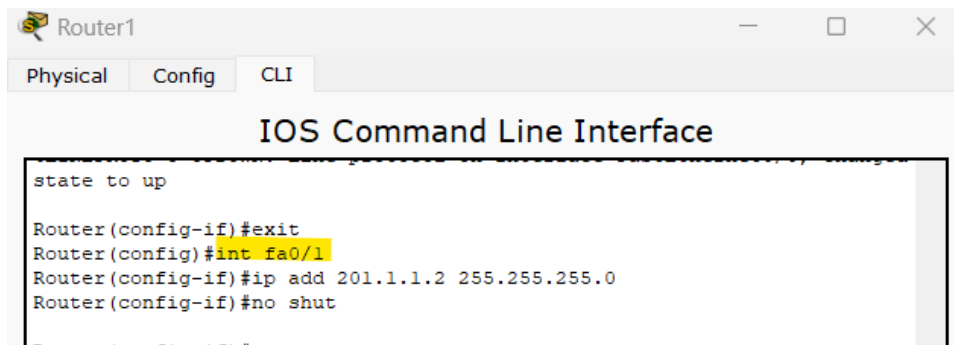
no shut – No shutdown command is used to activate the interfaces. It ensures that they are operational and able to transmit and receive data.



The screenshot shows a window titled 'Router0' with tabs for 'Physical', 'Config', and 'CLI'. The 'CLI' tab is active, displaying the 'IOS Command Line Interface'. The command history shows the following sequence: 'state to up', 'Router(config-if)#exit', 'Router(config)#int fa0/1', 'Router(config-if)#ip address 201.1.1.1 255.255.255.0', and 'Router(config-if)#no shut'. The command 'int fa0/1' is highlighted in yellow.

```
state to up
Router(config-if)#exit
Router(config)#int fa0/1
Router(config-if)#ip address 201.1.1.1 255.255.255.0
Router(config-if)#no shut
```

Figure 1.1.3 Router0 CLI Showing fa0/1 IP Address Configuration

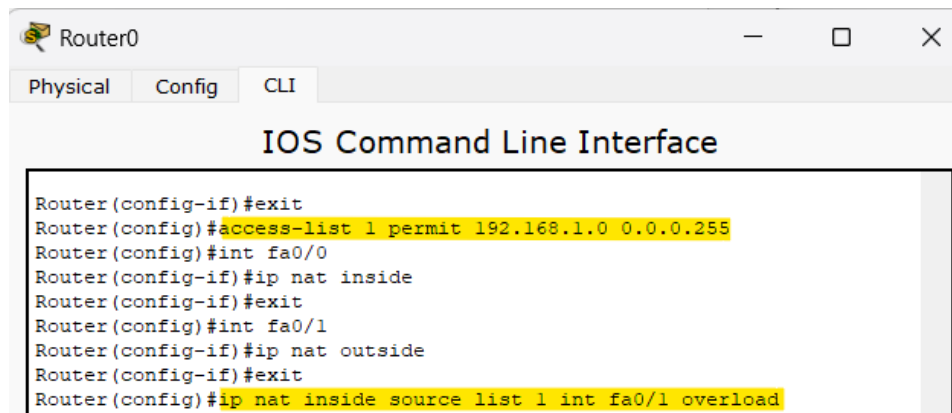


The screenshot shows a window titled 'Router1' with tabs for 'Physical', 'Config', and 'CLI'. The 'CLI' tab is active, displaying the 'IOS Command Line Interface'. The command history shows the following sequence: 'state to up', 'Router(config-if)#exit', 'Router(config)#int fa0/1', 'Router(config-if)#ip add 201.1.1.2 255.255.255.0', and 'Router(config-if)#no shut'. The command 'int fa0/1' is highlighted in yellow.

```
state to up
Router(config-if)#exit
Router(config)#int fa0/1
Router(config-if)#ip add 201.1.1.2 255.255.255.0
Router(config-if)#no shut
```

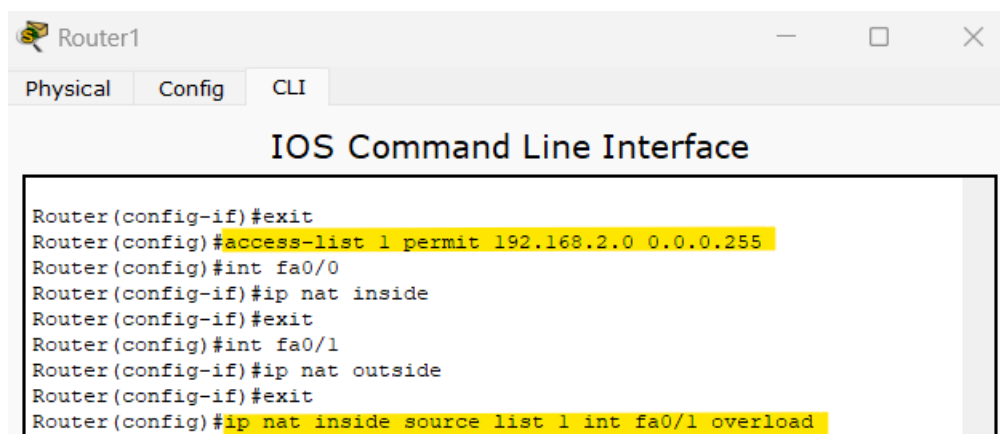
Figure 1.1.4 Router1 CLI Showing fa0/1 IP Address Configuration

int fa0/1 – Assigned the public address 201.1.1.1 for Router0 (*Figure 1.1.3*) and 201.1.1.2 for Router1 (*Figure 1.1.4*) which enables communication between the two routers and provide access to the external network including remote DHCP server.



```
Router0
Physical Config CLI
IOS Command Line Interface
Router(config-if)#exit
Router(config)#access-list 1 permit 192.168.1.0 0.0.0.255
Router(config)#int fa0/0
Router(config-if)#ip nat inside
Router(config-if)#exit
Router(config)#int fa0/1
Router(config-if)#ip nat outside
Router(config-if)#exit
Router(config)#ip nat inside source list 1 int fa0/1 overload
```

Figure 1.1.5 Router0 CLI Showing NAT Configuration for Internal Network



```
Router1
Physical Config CLI
IOS Command Line Interface
Router(config-if)#exit
Router(config)#access-list 1 permit 192.168.2.0 0.0.0.255
Router(config)#int fa0/0
Router(config-if)#ip nat inside
Router(config-if)#exit
Router(config)#int fa0/1
Router(config-if)#ip nat outside
Router(config-if)#exit
Router(config)#ip nat inside source list 1 int fa0/1 overload
```

Figure 1.1.6 Router1 CLI Showing NAT Configuration for Internal Network

Router(config)#access-list 1 permit 192.168.1.0 0.0.0.255

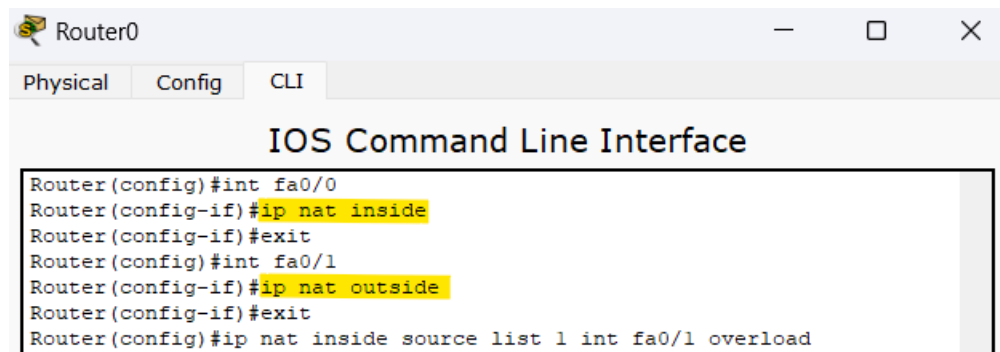
Router(config)#access-list 1 permit 192.168.2.0 0.0.0.255

An Access Control List (ACL 1) is created to define the internal networks on both routers 192.168.1.0/24 for Router0 (*Figure 1.1.5*) and 192.168.2.0/24 for Router1 (*Figure 1.1.6*). This ACL identifies which IP addresses can be translated by Network Address Translation (NAT).

Router(config)#ip nat inside source list 1 int fa0/1 overload

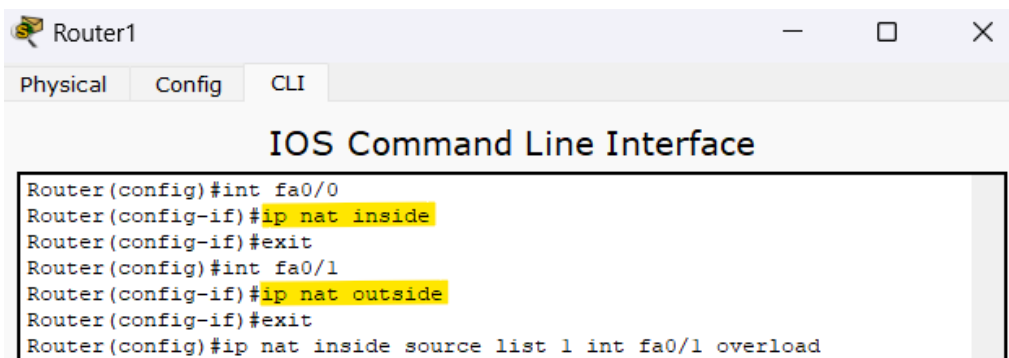
This command configures Port Address Translation (PAT) also called NAT overload. using the overload keyword (*Figure 1.1.6*). According to the lecture notes, PAT allows multiple internal hosts (PC0, PC1 & PC2) with private addresses to share a single public IP address assigned to the router outside interface (fa0/1). The NAT router tracks each session using unique source port

numbers. Thus, it enables all internal hosts to simultaneously access external networks while efficiently using limited public IPv4 addresses.

A screenshot of a network simulator window titled "Router0". It has tabs for "Physical", "Config", and "CLI", with "CLI" selected. The window displays the "IOS Command Line Interface". The command history shows: "Router(config)#int fa0/0", "Router(config-if)#ip nat inside", "Router(config-if)#exit", "Router(config)#int fa0/1", "Router(config-if)#ip nat outside", "Router(config-if)#exit", and "Router(config)#ip nat inside source list 1 int fa0/1 overload".

```
Router0
Physical Config CLI
IOS Command Line Interface
Router(config)#int fa0/0
Router(config-if)#ip nat inside
Router(config-if)#exit
Router(config)#int fa0/1
Router(config-if)#ip nat outside
Router(config-if)#exit
Router(config)#ip nat inside source list 1 int fa0/1 overload
```

Figure 1.1.7 Configuration of Internal and External NAT Interfaces on Router0

A screenshot of a network simulator window titled "Router1". It has tabs for "Physical", "Config", and "CLI", with "CLI" selected. The window displays the "IOS Command Line Interface". The command history shows: "Router(config)#int fa0/0", "Router(config-if)#ip nat inside", "Router(config-if)#exit", "Router(config)#int fa0/1", "Router(config-if)#ip nat outside", "Router(config-if)#exit", and "Router(config)#ip nat inside source list 1 int fa0/1 overload".

```
Router1
Physical Config CLI
IOS Command Line Interface
Router(config)#int fa0/0
Router(config-if)#ip nat inside
Router(config-if)#exit
Router(config)#int fa0/1
Router(config-if)#ip nat outside
Router(config-if)#exit
Router(config)#ip nat inside source list 1 int fa0/1 overload
```

Figure 1.1.8 Configuration of Internal and External NAT Interfaces on Router1

ip nat inside – The fa0/0 interface is configured as the internal (inside) interface, meaning it connects to the internal private network.

ip nat outside – The fa0/1 interface is configured as the external (outside) interface, meaning it connects to the external public network.

These commands specify the direction of traffic flow for NAT. This is important because NAT needs to know where private IP addresses come from and where they should be translated into public IP addresses. NAT wouldn't function properly without this setup.

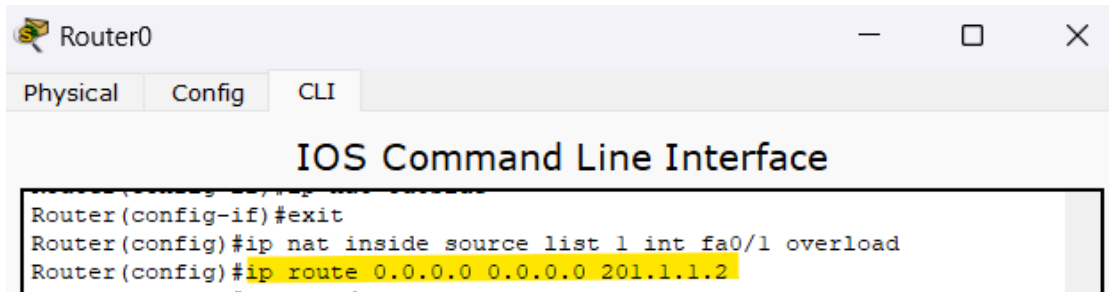


Figure 1.1.9 Default Route Configuration on Router0

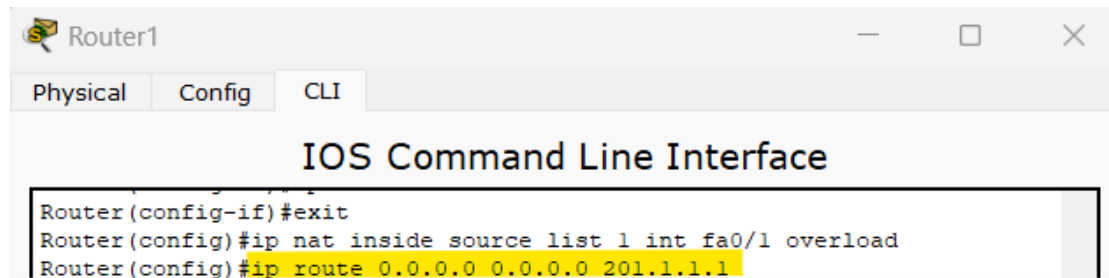


Figure 1.1.10 Default Route Configuration on Router1

This command configures a default route when the router does not have a specific route to a destination network.

Router(config)#ip route 0.0.0.0 0.0.0.0 201.1.1.2

On Router0, all unknown traffic is forwarded to the next-hop IP address 201.1.1.2 which is Router1 (*Figure 1.1.9*). This allows Router0 to send packets to external networks and ensures connectivity to the remote DHCP Server0.

Router(config)#ip route 0.0.0.0 0.0.0.0 201.1.1.1

On Router1, this command forwards all unknown traffic to Router0 (201.1.1.1) (*Figure 1.1.10*). It enables Router1 to reach the Company AA network (192.168.1.0/24) and other external services.

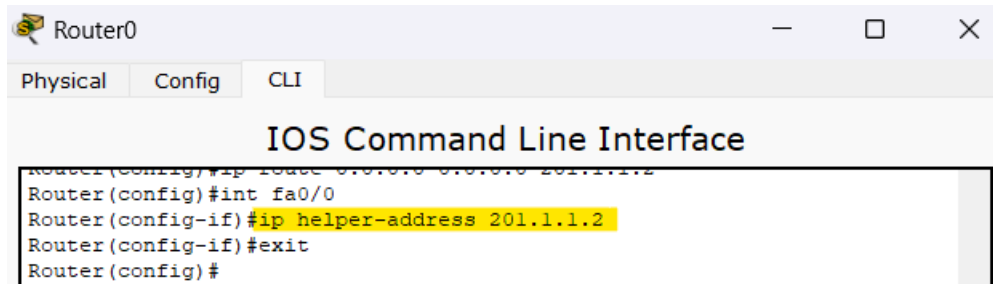


Figure 1.1.11 Configuration of DHCP Relay (`ip helper-address`) on Router0

Router(config-if)#ip helper-address 201.1.1.2

This command is a DHCP relay configuration. DHCP requests are broadcast messages and routers do not forward broadcasts by default. This means the devices in the 192.168.1.0/24 network cannot directly communicate with a DHCP server located in another subnet.

The `IP helper-address 201.1.1.2` command in Router0 (*Figure 1.1.11*) solves this problem by converting DHCP broadcast messages into unicast messages and forwarding them to the remote DHCP server. As a result, PC0, PC1 and PC2 can successfully obtain an IP address from the DHCP server in the offsite network (192.168.2.0/254).

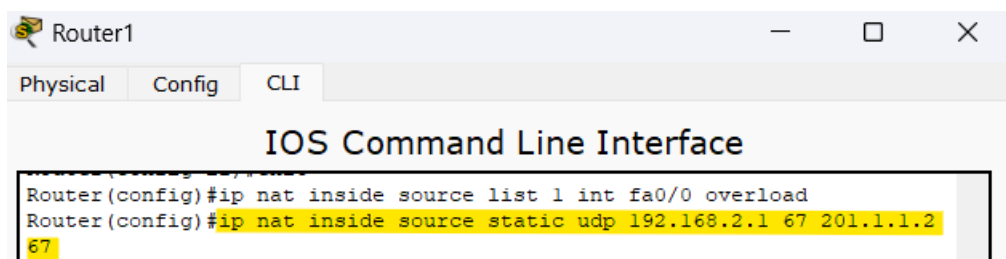


Figure 1.1.12 Configuration of Static NAT for DHCP Server (UDP Port 67) on Router1

Router(config)#ip nat inside source static udp 192.168.2.1 67 201.1.1.2 67

The static NAT rule for UDP port 67 is configured on Router1 to ensure that DHCP responses from the remote DHCP Server0 can successfully pass through the NAT and the internal network. Without this configuration, PCs in the 192.168.1.0/24 network would not be able to get an IP address from the DHCP server and would instead assign themselves an automatic APIPA address (169.254.x.x) such as the example as shown in *Figure 1.1.13*.

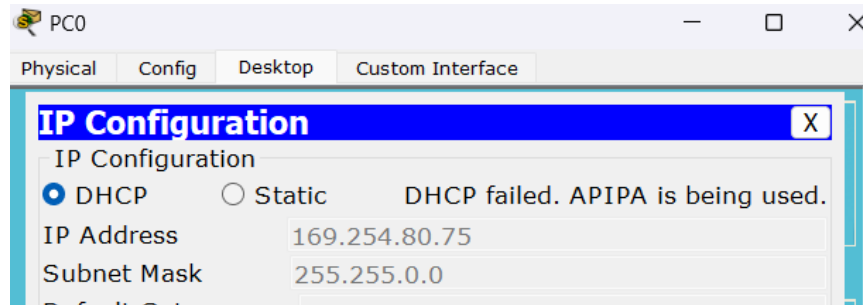


Figure 1.1.13 PC0 DHCP failure with APIPA address (169.254.80.75)

#Router0

```
Router > en
Router#conf t
Router(config)#int fa0/0
Router(config-if)#ip address 192.168.1.254 255.255.255.0
Router(config-if)#no shut
Router(config-if)#exit

Router(config)#int fa0/1
Router(config-if)#ip address 201.1.1.1 255.255.255.0
Router(config-if)#no shut
Router(config-if)#exit

Router(config)#access-list 1 permit 192.168.1.0 0.0.0.255
Router(config)#int fa0/0
Router(config-if)#ip nat inside
Router(config-if)#exit

Router(config)#int fa0/1
Router(config-if)#ip nat outside
Router(config-if)#exit
Router(config)#ip nat inside source list 1 interface fa0/1 overload

Router(config)#ip route 0.0.0.0 0.0.0.0 201.1.1.2

Router(config)#int fa0/0
Router(config-if)#ip helper-address 201.1.1.2
```

#Router 1

```
Router>en
Router#conf t
Router(config)#int fa0/0
Router(config-if)#ip address 192.168.2.254 255.255.255.0
Router(config-if)#no shut
Router(config-if)#no shut

Router(config)#int fa0/1
Router(config-if)#ip address 201.1.1.2 255.255.255.0
Router(config-if)#no shut
Router(config-if)#exit

Router(config)#access-list 1 permit 192.168.2.0 0.0.0.255
Router(config)#int fa0/0
Router(config-if)#ip nat inside
Router(config-if)#exit

Router(config)#int fa0/1
Router(config-if)#ip nat outside
Router(config-if)#exit
Router(config)#ip nat inside source list 1 int fa0/1 overload

Router(config)#ip nat inside source static udp 192.168.2.1
67 201.1.1.2 67

Router(config)#ip route 0.0.0.0 0.0.0.0 201.1.1.1
```

(b) Getting Address from DHCP

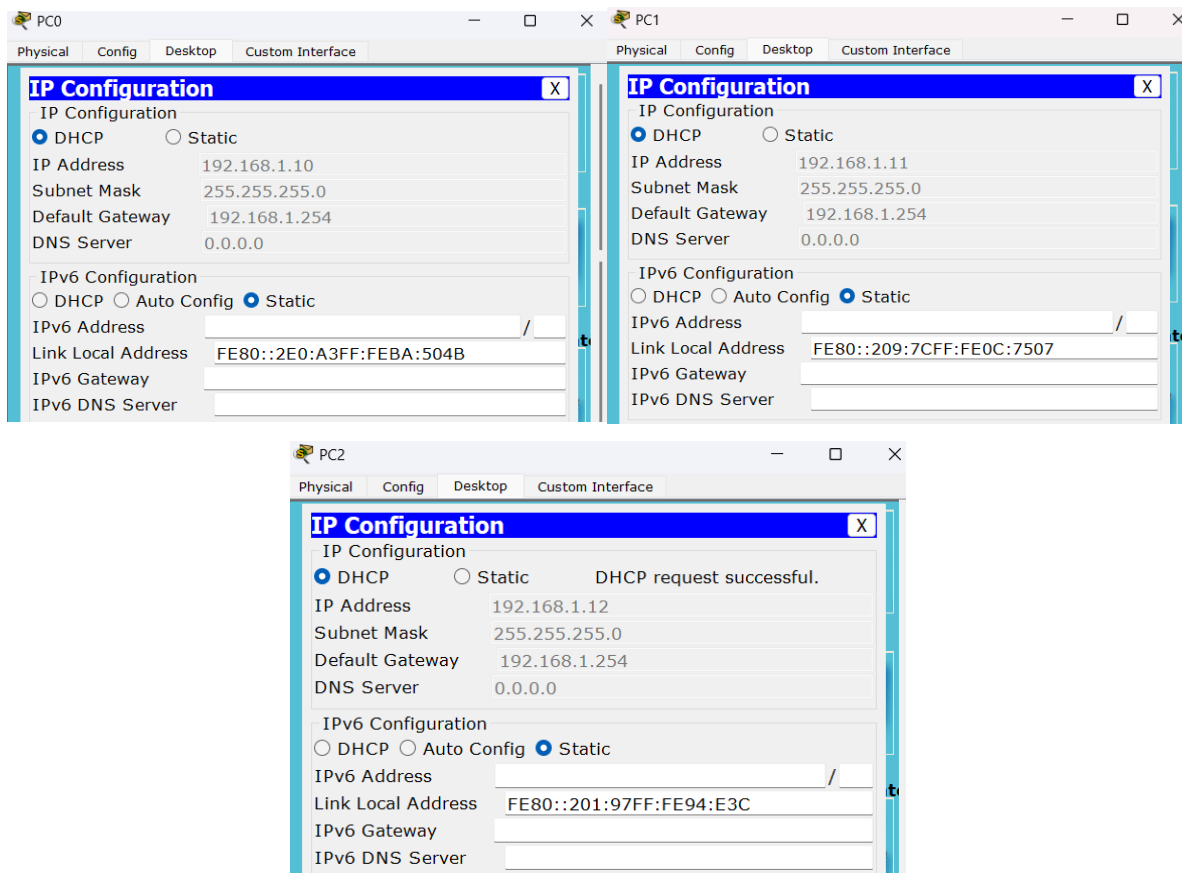


Figure 1.2.1 DHCP Configuration on PC0-PC2 with Obtained IP Addresses

DHCP is utilized to dynamically assign IP addresses to devices on a network for communication purposes without manual configuration [1]. Based on the configurations, all PCs (PC0, PC1 and PC2) have successfully obtained their respective IP address from the Remote DHCP Server using '**ip helper-address**' command configured on Router0. This command enables the router to act as a middleman that help forwards DHCP requests to more centralized DHCP servers located on a different network [2].

Normally, when a PC attempt to get an IP address, the **DORA process** (Discover, Offer, Request, Acknowledge) is initiated where it sends a **DHCP Discover message** as a broadcast to all DHCP servers within range to locate available DHCP servers [3]. However, broadcast messages are limited to LAN and usually cannot pass through routers.

With the **ip helper-address** configured, Router0 detects a request and intercepts the broadcast **DHCP request** and converts the broadcast message into unicast message and forwards it directly to the Remote DHCP Server.

Once DHCP receives the request, the DHCP server selects an available IP address from its pool and responds with a **DHCP Offer message** as a unicast packet back to Router0. The router will then forward this response to the requesting PC within the LAN. The PC receives the offer and then replies with a **DHCP Request** to confirm the offered IP address. This request again reaches DHCP through the help of helper address in Router0. Finally, the DHCP server sends a **DHCP Acknowledge message** to complete the process.

As a result, all PCs (PC0, PC1 and PC2) will receive an IP address within a valid range of their network address even though the DHCP server is on a different network as shown in *Figure 1.2.1*.

(c) Pinging DHCP Server

Fire	Last Status	Source	Destination	Type	Color	Time(se	Periodic	Num	Edit	Delete
	Failed	PC0	Remote DH...	ICMP		0.000	N	0	(edit)	(delete)
	Failed	PC0	Remote DH...	ICMP		0.000	N	1	(edit)	(delete)
	Failed	PC0	Remote DH...	ICMP		0.000	N	2	(edit)	(delete)

Figure 1.3.1 Failed ICMP Communication from PC0 to Remote DHCP Server

No, the PCs (PC0, PC1 and PC2) are unable to directly ping the Remote DHCP server as shown in *Figure 1.3.1*. Even though PCs successfully obtain their IP addresses from the DHCP server, it does not necessarily mean it can ping the server. In this case, Port Address Translation (PAT) is configured on both routers which is useful for internet access. However, internal communication between the two branches becomes complicated.

When an ICMP packet sent from a PC reaches Router0, Router0 will translates the private source IP address to its public IP address (201.1.1.1) by using NAT before the packet over the public network to Router1. When Router1 receives the packet, it translates the destination IP from its public IP (201.1.1.2) back to the private IP (192.168.2) and forwards it toward the DHCP server on its internal network.

However, the issue arises during the return path. When the DHCP server receives the ICMP packet, it sees the source IP as 201.1.1.1 (Router0's public IP) rather than the original private IP of the PC. The server sends it reply toward Router0 via Router1. Due to NAT behaviour and the absence of a proper translation entry or matching session for ICMP, the returning packet cannot be correctly mapped back to the original internal host (PC0, PC1, or PC2).

Additionally, Router1 detects that the destination IP (201.1.1.1) is directly connected to its own interface. Because there is no NAT translation for ICMP, the router cannot forward the reply to the originating PC that results in the packet being dropped. This means that while static NAT allows DHCP requests to reach the server, ICMP replies fail to return to the LAN hosts.

As a result, the ICMP reply never reaches the originating PC and causing the ping to fail, even though DHCP requests succeed due to static NAT on UDP port 67.

Part 2

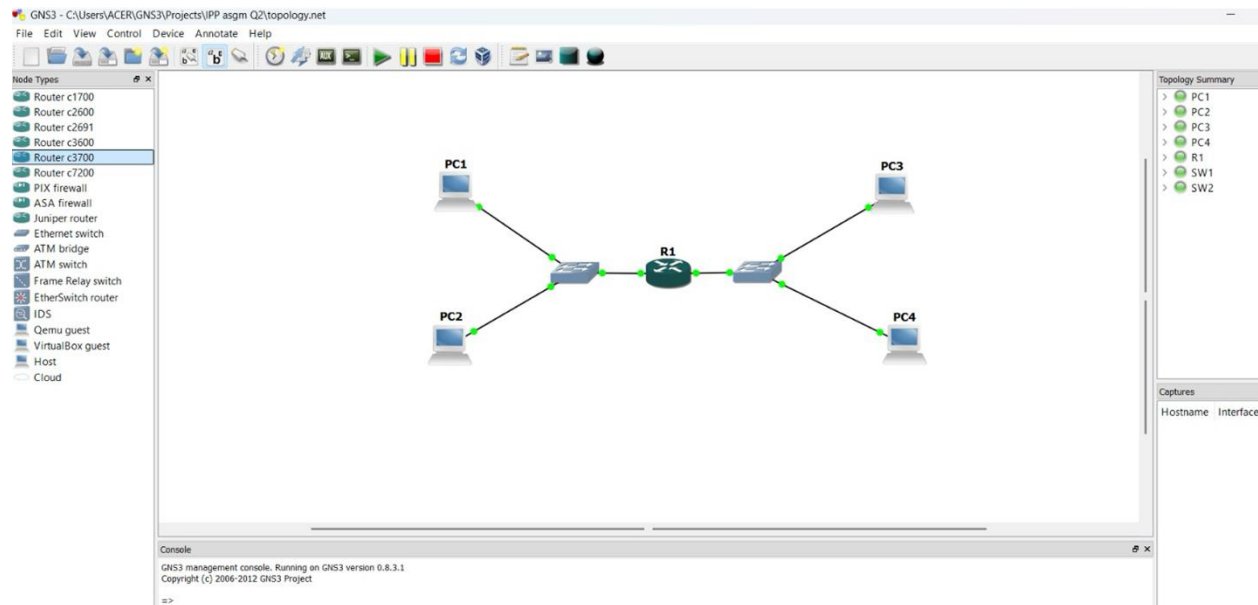


Figure 2.0.1 Network Topology Diagram

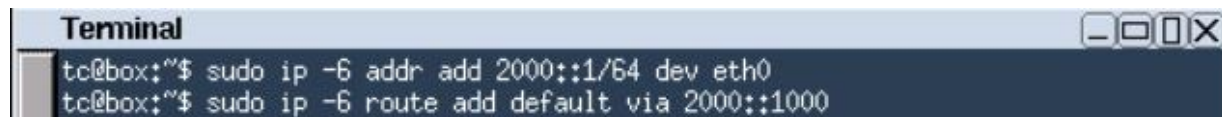
In this setup, we use GNS3 to build the network topology, where a router using the c3745 image connects two Local Area Networks (LANs) through Ethernet switches. The end devices PC1, PC2, PC3 and PC4 are implemented using QEMU Microcore images. QEMU is chosen because it is a machine emulator that allows an unmodified operating system and all its applications to run within a virtual machine [4]. This capability enables the use of real Linux networking commands. Additionally, QEMU can simulate NE2000 and other generic network cards [5] and can provide realistic packet headers for Wireshark analysis. All links connecting the PCs to switches and switches to the router are configured as Fast Ethernet links.

```
Terminal
tc@box:~$ sudo ifconfig eth0 192.168.1.1 netmask 255.255.255.0 up
tc@box:~$ sudo route add default gw 192.168.1.254
```

Figure 2.0.2 Interface and Routing Configuration on PC1

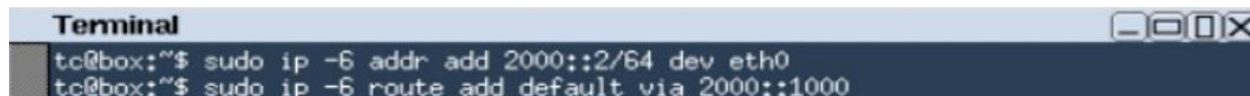
```
Terminal
tc@box:~$ sudo ifconfig eth0 192.168.1.2 netmask 255.255.255.0 up
tc@box:~$ sudo route add default gw 192.168.1.254
```

Figure 2.0.3 Interface and Routing Configuration on PC2

A terminal window titled "Terminal" with standard window controls. It shows two lines of command execution: `tc@box:~$ sudo ip -6 addr add 2000::1/64 dev eth0` and `tc@box:~$ sudo ip -6 route add default via 2000::1000`.

```
Terminal
tc@box:~$ sudo ip -6 addr add 2000::1/64 dev eth0
tc@box:~$ sudo ip -6 route add default via 2000::1000
```

Figure 2.0.4 Interface and Routing Configuration on PC3

A terminal window titled "Terminal" with standard window controls. It shows two lines of command execution: `tc@box:~$ sudo ip -6 addr add 2000::2/64 dev eth0` and `tc@box:~$ sudo ip -6 route add default via 2000::1000`.

```
Terminal
tc@box:~$ sudo ip -6 addr add 2000::2/64 dev eth0
tc@box:~$ sudo ip -6 route add default via 2000::1000
```

Figure 2.0.5 Interface and Routing Configuration on PC4

Each LAN is configured with a different IP protocol version. The left-hand LAN containing PC1 and PC2 uses IPv4 addresses 192.168.1.1 and 192.168.1.2 with subnet mask 255.255.255.0. The right-hand LAN containing PC3 and PC4 uses IPv6 addresses 2000::1 and 2000::2 with a /64 prefix. These subnet masks define their respective network ranges.

The router supports both protocols. Default gateway addresses are configured to enable communication beyond the local network. PC1 and PC2 use 192.168.1.254 as their default gateway, while PC3 and PC4 use 2000::1000 as their default gateway, which corresponds to the router's respective IPv4 and IPv6 interfaces. This configuration ensures that all devices can communicate with the router, which is essential for enabling inter-network communication and supporting NAT-PT translation between IPv4 and IPv6 networks.

(a) Static NAT-PT Mapping and Protocol Translation Configuration

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ipv6 unicast-routing
R1(config)#int fa0/0
R1(config-if)#ip address 192.168.1.254 255.255.255.0
R1(config-if)#ipv6 nat
R1(config-if)#n
*Mar  1 00:36:50.547: %LINEPROTO-5-UPDOWN: Line protocol on Interface NVI0, changed state to up
R1(config-if)#no shut
R1(config-if)#
*Mar  1 00:37:01.423: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar  1 00:37:02.423: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R1(config-if)#exit
R1(config)#int fa0/1
R1(config-if)#ipv6 address 2000::1000/64
R1(config-if)#ipv6 nat
R1(config-if)#no shut
R1(config-if)#
*Mar  1 00:37:56.279: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar  1 00:37:57.279: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R1(config-if)#exit
```

Figure 2.1.1 IPv4 and IPv6 Configuration on R1

The configuration shown sets up a router to support communication between an IPv4 network and an IPv6 network using Network Address Translation-Protocol Translation (NAT-PT). The process begins with the command **ipv6 unicast-routing**, which is essential because it enables the router to forward IPv6 packets. In addition, this command activates IPv6 functionality on the device by creating an IPv6 routing table, allowing IPv6 addresses to be configured on interfaces and enabling access to various IPv6 features and services supported by the router [6]. Communication across networks would be impossible without this command as the router would act like a normal host and would not route any IPv6 traffic between interfaces.

Next, the configuration moves to the FastEthernet0/0 interface using **int fa0/0**. This interface is configured with the **ip address 192.168.1.254 255.255.255.0**, which serves as the default gateway for all devices in the IPv4 LAN such as PC1 and PC2. Any IPv4 packets from these devices destined for external networks will be sent to this address. The command **ipv6 nat** is then applied to this interface, which enables the router to perform protocol translation on packets entering or leaving this interface. This is important because it allows IPv4 packets to be translated into IPv6 format when communicating with devices on the IPv6 network. The **no shutdown** command is used to activate the interface, ensuring that it is operational and able to transmit and receive data. Without this command, the interface would remain administratively down and unusable.

After configuring the IPv4 side, the setup proceeds to the FastEthernet0/1 interface using **int fa0/1**. This interface is assigned the IPv6 address 2000::1000/64, which acts as the default gateway for devices in the IPv6 LAN such as PC3 and PC4. Devices in this network will send their outgoing traffic to this address when attempting to communicate outside their local subnet. Similar to the IPv4 interface, the **ipv6 nat** command is applied here to enable NAT-PT processing. This allows IPv6 packets to be translated into IPv4 formats when communicating with devices on the IPv4 network. The **no shutdown** command is again used to bring the interface up, ensuring that the IPv6 network is active and reachable.

```
R1(config)#ipv6 nat prefix 2001::/96
R1(config)#ipv6 nat v4v6 source 192.168.1.1 2001::1
R1(config)#ipv6 nat v4v6 source 192.168.1.2 2001::2
R1(config)#ipv6 nat v6v4 source 2000::1 192.168.2.1
R1(config)#ipv6 nat v6v4 source 2000::2 192.168.2.2
*Mar  1 00:40:16.403: % Address 192.168.2.1 already in use on attached network
R1(config)#ipv6 nat v6v4 source 2000::2 192.168.2.2
R1(config)#
*Mar  1 00:40:26.895: % Address 192.168.2.2 already in use on attached network
```

Figure 2.1.2 Configuration of Static NAT-PT Mappings on R1

The command **ipv6 nat prefix 2001::/96** specifies a well-known IPv6 prefix that is used to represent IPv4 addresses in IPv6 format. In NAT-PT, IPv4 addresses are embedded into an IPv6 address and the /96 prefix means the first 96 bits are fixed while the remaining 32 bits are used to carry the IPv4 address [7]. This prefix acts as a bridge between the two protocols, enabling the router to algorithmically translate IPv4 addresses into IPv6 equivalents and vice versa.

The commands **ipv6 nat v4v6 source 192.168.1.1 2001::1** and **ipv6 nat v4v6 source 192.168.1.2 2001::2** create static mappings from IPv4 to IPv6 addresses. When an IPv4 device such as 192.168.1.1 sends packets toward the IPv6 network, the router translates its source address into the corresponding IPv6 address 2001::1. Similarly, 192.168.1.2 is always translated to 2001::2.

On the other hand, the commands **ipv6 nat v6v4 source 2000::1 192.168.2.1** and **ipv6 nat v6v4 source 2000::2 192.168.2.2** define static mappings from IPv6 to IPv4 addresses. When IPv6 hosts such as PC3 send packets to the IPv4 network, its IPv6 address 2000::1 is translated into IPv4 address 192.168.2.1. This allows IPv6-only devices to communicate with IPv4-only devices, with the router handling the protocol conversion transparently.

In this configuration, static NAT is used to provide fixed, one-to-one address mappings between IPv4 and IPv6 hosts, ensuring consistent and predictable communication. Static NAT

maps network traffic from a static external IP address to an internal IP address by creating permanent translations between real and mapped addresses [8]. In this case, the IPv4 addresses 192.168.1.1 and 192.168.1.2 represent internal private hosts, while their corresponding IPv6 addresses 2001::1 and 2001::2 act as the mapped external representations in the IPv6 network. Similarly, Ipv6 hosts 2000::1 and 2000::2 are statically mapped to IPv4 addresses 192.168.2.1 and 192.168.2.2. These fixed mappings ensure that each device always uses the same translated address, allowing reliable bidirectional communication.

```
R1#show ipv6 nat translation
```

Prot	IPv4 source	IPv6 source
	IPv4 destination	IPv6 destination
---	---	---
	192.168.1.1	2001::1
---	---	---
	192.168.1.2	2001::2
---	192.168.2.1	2000::1
	---	---
---	192.168.2.2	2000::2
	---	---

Figure 2.1.3 NAT-PT Translation Table

The **show ipv6 nat translation** table confirms that the NAT-PT configuration is functioning correctly. It displays the established static mapping between IPv4 and Ipv6 addresses, indicating that translation entries are successfully created on the router. The presence of both IPv4-to-IPv6 and IPv6-to-IPv4 mappings shows that bidirectional communication is properly supported. This verifies that the configuration has been implemented correctly and that devices from both networks can communicate through consistent address translation.

Full commands:

```
R1# conf t
```

```
R1(config)# ipv6 unicast-routing
```

```
R1(config)# interface FastEthernet0/0
```

```
R1(config-if)# ip address 192.168.1.254 255.255.255.0
```

```
R1(config-if)# ipv6 nat
```

```
R1(config-if)# no shutdown
R1(config-if)# exit
```

```
R1(config)# interface FastEthernet0/1
R1(config-if)# ipv6 address 2000::1000/64
R1(config-if)# ipv6 nat
R1(config-if)# no shutdown
R1(config-if)# exit
```

```
R1(config)# ipv6 nat prefix 2001::/96
R1(config)# ipv6 nat v4v6 source 192.168.1.1 2001::1
R1(config)# ipv6 nat v4v6 source 192.168.1.2 2001::2
R1(config)# ipv6 nat v6v4 source 2000::1 192.168.2.1
R1(config)# ipv6 nat v6v4 source 2000::2 192.168.2.2
```

(b) Running Configuration on R1

```
R1
R1#show running-config
Building configuration...

Current configuration : 1084 bytes
!
version 12.4
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname R1
!
boot-start-marker
boot-end-marker
!
no aaa new-model
memory-size iomem 5
ip cef
!
!
!
no ip domain lookup
ip domain name lab.local
ip auth-proxy max-nodata-conns 3
ip admission max-nodata-conns 3
!
ip v6 unicast-routing
```

Figure 2.2.1 Running Configuration part 1


```

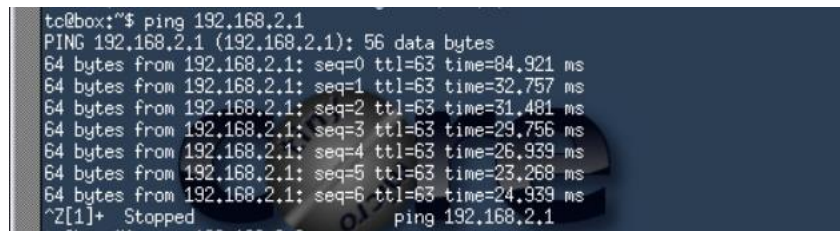
R1
interface FastEthernet0/0
 ip address 192.168.1.254 255.255.255.0
 duplex auto
 speed auto
 ipv6 nat
!
interface FastEthernet0/1
 no ip address
 duplex auto
 speed auto
 ipv6 address 2000::1000/64
 ipv6 nat
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
!
ipv6 nat v4v6 source 192.168.1.1 2001::1
ipv6 nat v4v6 source 192.168.1.2 2001::2
ipv6 nat v6v4 source 2000::1 192.168.2.1
ipv6 nat v6v4 source 2000::2 192.168.2.2
ipv6 nat prefix 2001::/96
!
!
!
control-plane
!
!
!
!
!
!
!
!
!
!
line con 0
 exec-timeout 0 0
 privilege level 15
 logging synchronous
line aux 0
 exec-timeout 0 0
 privilege level 15
 logging synchronous
line vty 0 4
 login
!

```

Figure 2.2.2 Running Configuration part 2

The running configuration of Router R1 is designed to support both IPv4 and IPv6 networking while enabling translation between the two using NAT. IPv6 routing is enabled globally using `ipv6 unicast-routing`, allowing the router to forward IPv6 packets. Interface FastEthernet0/0 is configured with an IPv4 address 192.168.1.254/24, serving as the gateway for the IPv4 LAN, while FastEthernet0/1 is configured with an IPv6 address 2000::1000/64, acting as the IPv6-facing interface. Both interfaces have `ipv6 nat` enabled, indicating they participate in NAT translation. The router is configured with a NAT64 prefix 2001::/96, which is used to map IPv4 addresses into IPv6 format. Static NAT mappings are defined for both directions. v4v6 mappings translate specific IPv4 addresses 192.168.1.1 and 192.168.1.2 into IPv6 addresses 2001::1 and 2001::2, while v6v4 mappings translate IPv6 addresses 2000::1 and 2000::2 into IPv4 addresses 192.168.2.1 and 192.168.2.2. Overall, this configuration enables communication between IPv4-only and IPv6-only hosts through bidirectional NAT translation.

(c) Verification of Bidirectional Connectivity via NAT-PT

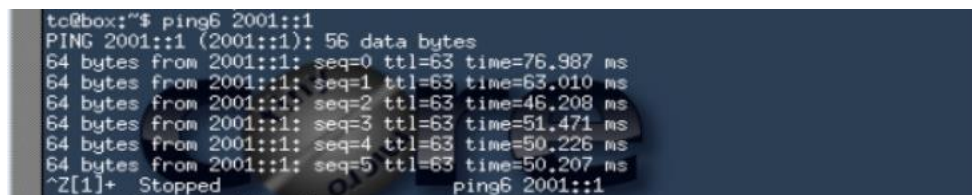


```
tc@box1:~$ ping 192.168.2.1
PING 192.168.2.1 (192.168.2.1): 56 data bytes
64 bytes from 192.168.2.1: seq=0 ttl=63 time=84.921 ms
64 bytes from 192.168.2.1: seq=1 ttl=63 time=32.757 ms
64 bytes from 192.168.2.1: seq=2 ttl=63 time=31.481 ms
64 bytes from 192.168.2.1: seq=3 ttl=63 time=29.756 ms
64 bytes from 192.168.2.1: seq=4 ttl=63 time=26.939 ms
64 bytes from 192.168.2.1: seq=5 ttl=63 time=23.268 ms
64 bytes from 192.168.2.1: seq=6 ttl=63 time=24.939 ms
^Z[1]+ Stopped ping 192.168.2.1
```

Figure 2.3.1 Pinging from IPv4 (PC1) to IPv6 (PC3)

When PC1 or PC2 initiates a ping towards PC3 or PC4, communication is enabled through NAT translation configured on R1. Figure 2.3.1 illustrates a successful ping from PC1 to PC3. When PC1 (192.168.1.1) sends an ICMP Echo Request, the packet is originally an IPv4 packet with the destination set to PC3's IPv4 address (192.168.2.1). This packet is first forwarded to its default gateway, R1 (192.168.1.254). Based on the configured `ipv6 nat v4v6` rule, R1 translates the IPv4 source address (192.168.1.1) into an IPv6 address (2001::1) and also converts the IPv4 destination address into its corresponding IPv6-mapped address using the NAT64 prefix (2001::/96). The packet is therefore transformed into an IPv6 packet before being forwarded out through FastEthernet0/1 toward the IPv6 network.

As the packet is processed for delivery to the destination network, R1 applies the `ipv6 nat v6v4` rule, translating the IPv6 destination address back into the original IPv4 address (192.168.2.1) of PC3. Thus, PC3 ultimately receives a standard IPv4 packet. The reply generated by PC3 follows the reverse process. It is sent as an IPv4 Echo Reply, translated into IPv6 when entering R1, and then translated back into IPv4 before reaching PC1. The successful exchange of Echo Request and Reply messages confirms that bidirectional NAT translation is functioning correctly, enabling seamless communication between IPv4 hosts and hosts across the translated network.



```
tc@box1:~$ ping6 2001::1
PING 2001::1 (2001::1): 56 data bytes
64 bytes from 2001::1: seq=0 ttl=63 time=76.987 ms
64 bytes from 2001::1: seq=1 ttl=63 time=63.010 ms
64 bytes from 2001::1: seq=2 ttl=63 time=46.208 ms
64 bytes from 2001::1: seq=3 ttl=63 time=51.471 ms
64 bytes from 2001::1: seq=4 ttl=63 time=50.226 ms
64 bytes from 2001::1: seq=5 ttl=63 time=50.207 ms
^Z[1]+ Stopped ping6 2001::1
```

Figure 2.3.2 Pinging from IPv6 (PC3) to IPv4 (PC1)

When PC3 or PC4 initiates a ping toward PC1 or PC2, the communication process occurs in the reverse direction using the same NAT translation rules configured on R1. Figure 2.3.2 illustrates a successful ping from PC3 to PC1. When PC3 sends an ICMP Echo Request, the packet is generated as an IPv6 packet, with the source address (2000::1) and the destination set to the IPv6-mapped address representing PC1 (2001::1), rather than the native IPv4 address. This packet is forwarded to R1, where the ipv6 nat v6v4 rule is applied. At this stage, R1 translates the IPv6 source address into its corresponding IPv4 address (192.168.2.1) and simultaneously converts the IPv6 destination address (2001::1) into the actual IPv4 address of PC1 (192.168.1.1).

The packet is thus transformed into a standard IPv4 packet and forwarded out through FastEthernet0/0 toward the IPv4 network. Upon receiving the packet, PC1 processes it as a normal ICMP Echo Request and generates an Echo Reply, which is sent back to its default gateway, R1. R1 then performs the reverse translation by converting the IPv4 packet into IPv6 format, mapping both the source and destination addresses accordingly, before forwarding it back to PC3.

(d) Wireshark Protocol Analysis: Header Translation Evidence

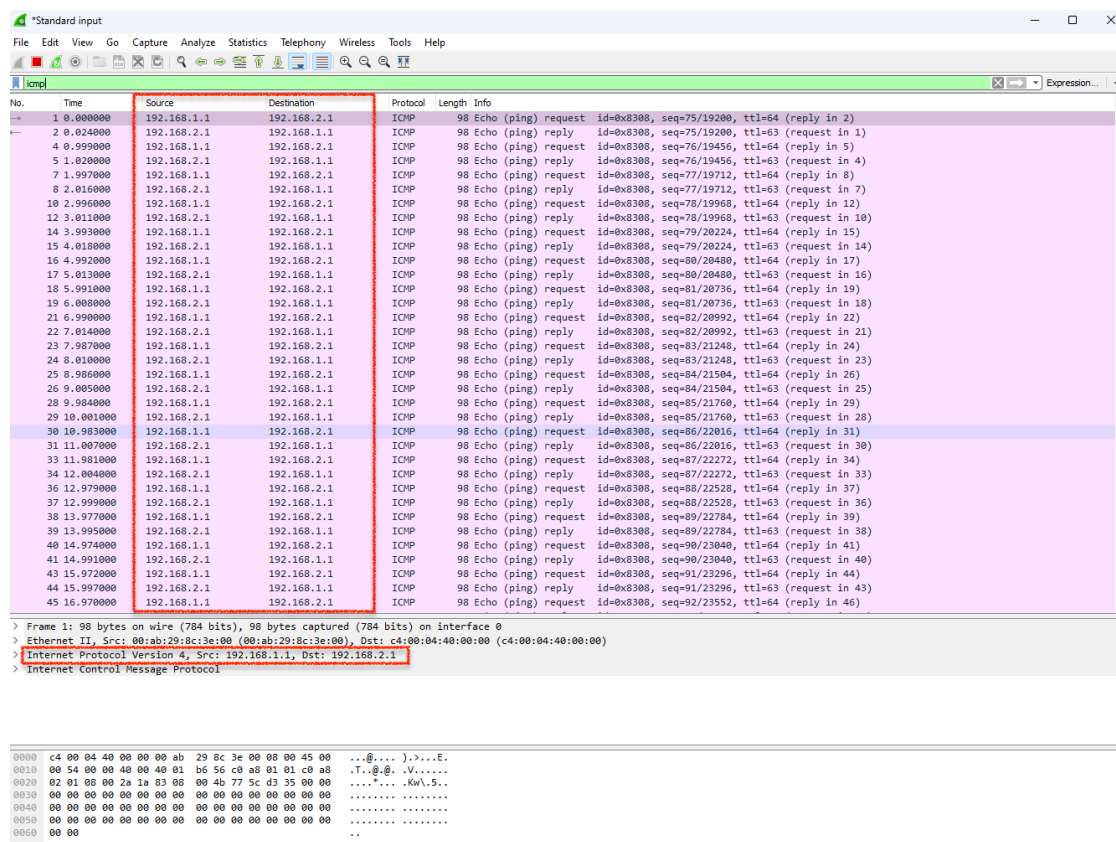


Figure 2.4.1 ICMPv4 Packet Capture on the PC1-R1 Link Segment

The Wireshark captures obtained from the GNS3 environment provide clear evidence of the bidirectional communication process. The packet list pane shows multiple ICMP Echo Request and Reply packets exchanged between 192.168.1.1 and the virtual address 192.168.2.1, demonstrating successful connectivity from the perspective of the IPv4 hosts. As highlighted in the packet details pane, the selected frame is explicitly identified as “Internet Protocol Version 4”. This confirms that the traffic remains in its native IPv4 format on this segment of the network before reaching R1, where the NAT64 translation engine reconstructs the headers into IPv6 for transit to the destination network.

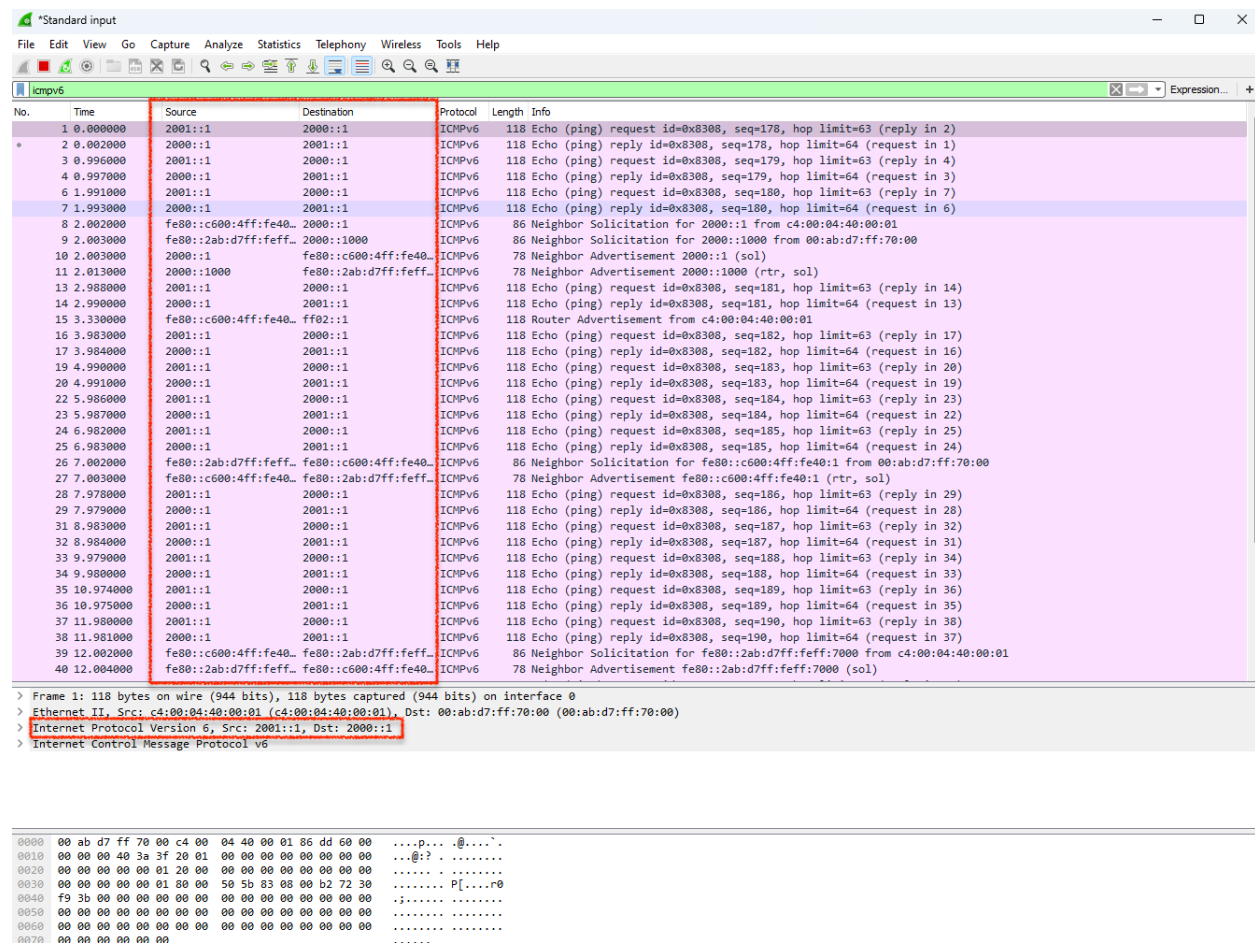


Figure 2.4.2 Translated ICMPv6 Packet Capture on the R1-PC3 Link Segment

The packet list pane in this capture confirms the successful execution of the translation process, displaying ICMPv6 Echo Request and Reply packets alongside Neighbor Discovery Protocol (NDP) messages. The addressing scheme utilizes Global Unicast Addresses (GUA) such

as 2000::1 and Link-Local addresses within the fe80::/10 prefix, which are essential for local segment operations like MAC address resolution. Upon selecting a packet, the packet details pane explicitly identifies the protocol as “Internet Protocol Version 6”, presenting the reconstructed IPv6 source and destination headers. By comparing this to the previous IPv4 capture, it is evident that the translation engine is actively performing header conversion, successfully remapping the entire packet structure to maintain bidirectional connectivity between the two disparate protocol stacks.

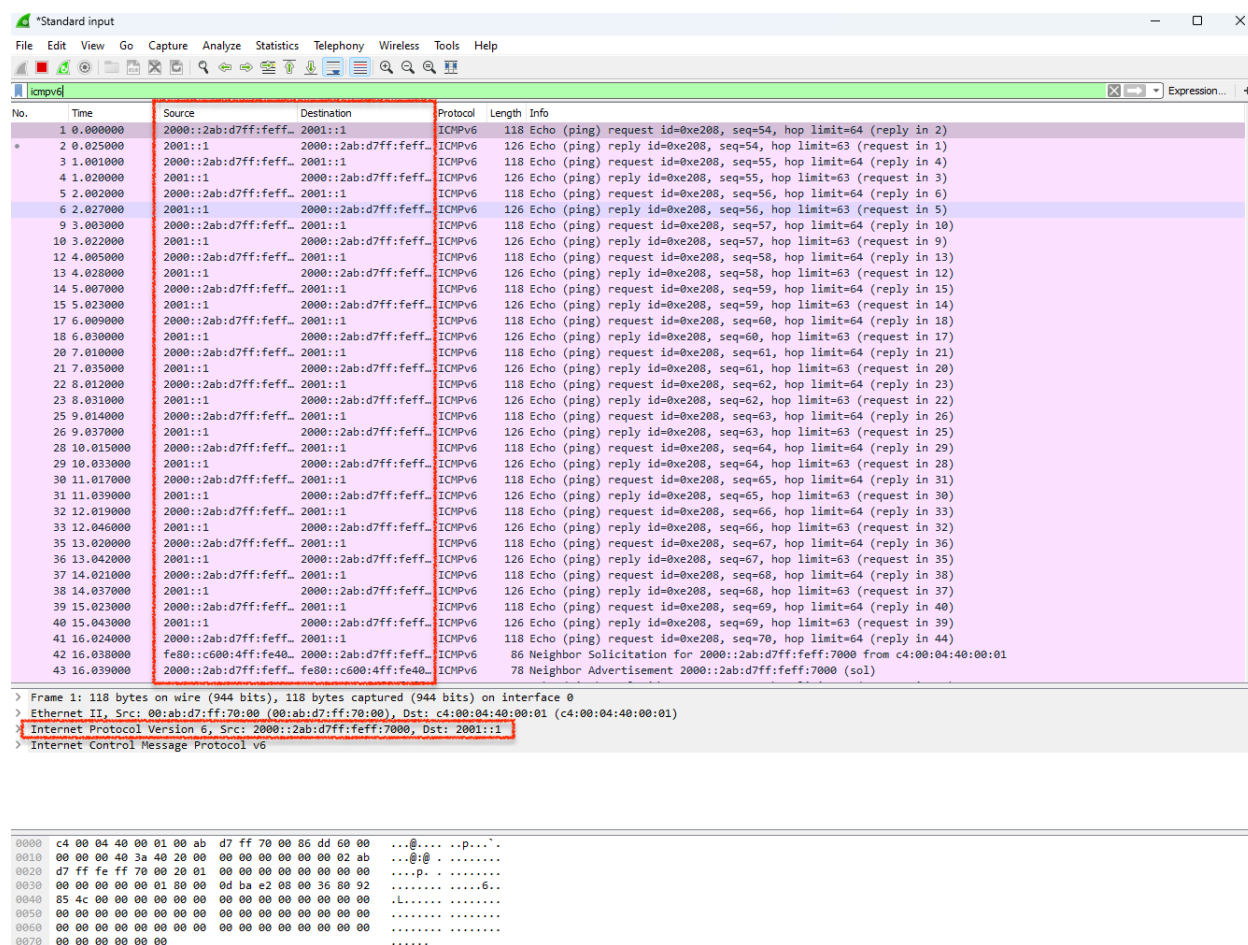


Figure 2.4.3 ICMPv6 Packet Capture on the R1-PC4 Link Segment

When communication is initiated from the IPv6 segment, the Wireshark capture first displays ICMPv6 Echo Request packets originating from PC4’s global unicast address (2000::2) and destined for the translated IPv6 address representing PC1 (2001::1). In the packet list pane, this is observed as native ICMPv6 traffic, accompanied by Neighbor Discovery Protocol

(NDP) messages which facilitate address resolution on the IPv6 link. Upon selecting a specific packet, the packet details pane explicitly identifies the protocol as "Internet Protocol Version 6," confirming that the session originates within the IPv6 domain before any translation occurs. This capture serves as the baseline evidence for the reverse translation path, showing the packet's state before the NAT64 translation engine converts the headers for the IPv4 destination.

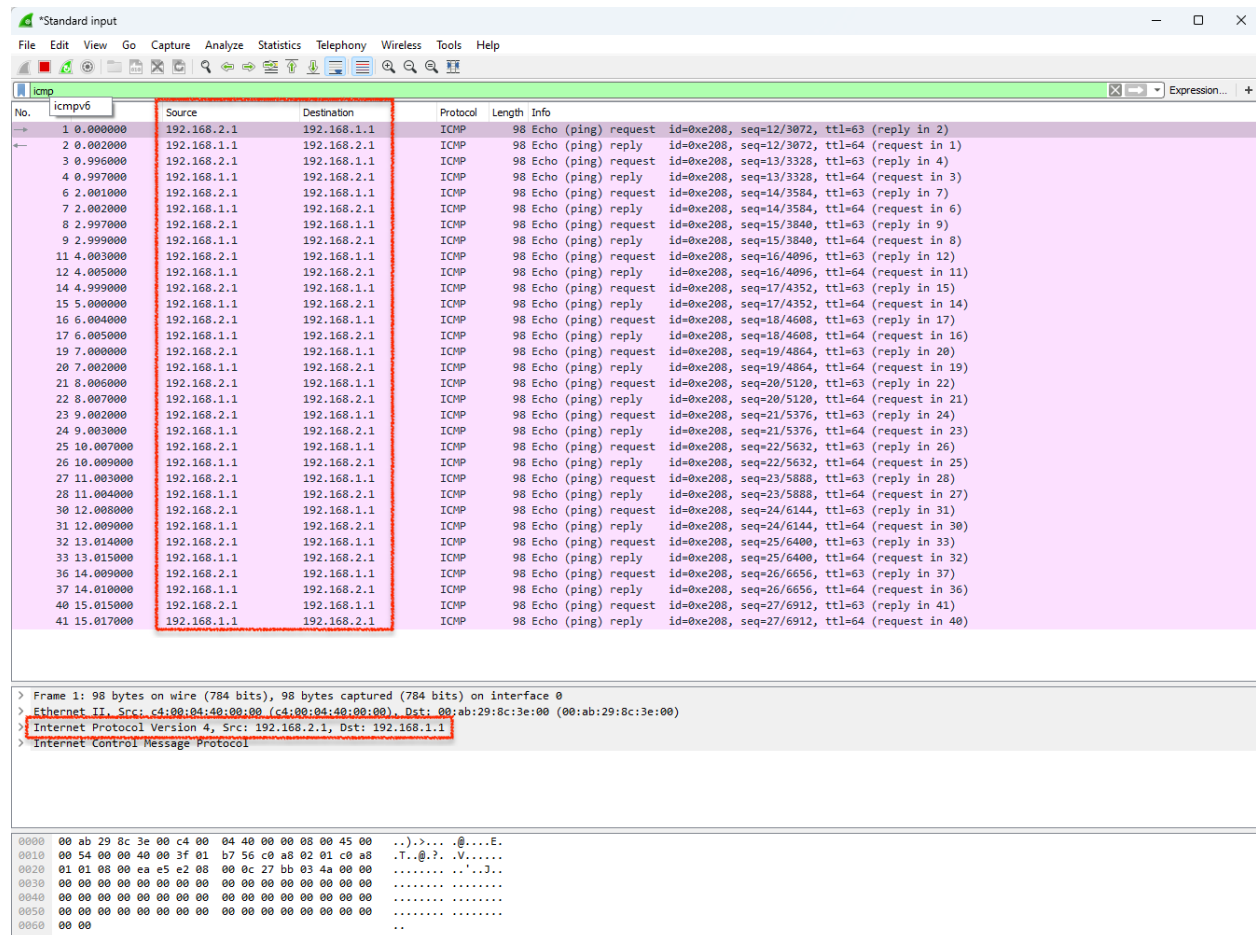


Figure 2.4.4 Translated ICMPv4 Packet Capture on the PC1-R1 Link Segment

As the packet traverses the translation device, the NAT64 translation engine converts the IPv6 header into a standard IPv4 header before it reaches the destination segment. In the corresponding Wireshark capture on the IPv4 side, this traffic is identified as ICMP Echo Request packets with a source of 192.168.2.1 and a destination of 192.168.1.1. PC1 then generates an IPv4 Echo Reply, which the router subsequently translates back into ICMPv6 format using the predefined static mappings. This return traffic is visible in the IPv6 capture as ICMPv6 Echo

Reply packets, confirming that bidirectional protocol translation is fully functional and that the router is successfully managing the session state between the two disparate IP versions.

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